

Zhengqi Hou

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Skills

Programming Languages: C/C++, Python, Java, PHP, SQL, JavaScript, TypeScript, OCaml, Assembly

Technologies: Linux, Windows, MacOS, MySQL Workbench, PostgreSQL, SQLAlchemy, AWS Services (EC2, S3, Lambda, RDS), React.js, Vite, Express.js, RESTful API, FastAPI, Firebase, Nginx, Unreal Engine, PyTorch, OpenGL, LLM, ThinkPHP

Education

Northeastern University | San Jose, California

Sep. 2025 – May 2027

Master of Science in Artificial Intelligence

University of Illinois Urbana-Champaign | Champaign, Illinois

May 2025

Bachelor of Science in Computer Science

GPA: 3.96/4.0

- **Coursework:** Data Structures, Algorithms and Models of Computation, Database Systems, System Programming, Cloud Networking, Data Management in the Cloud, Game Development, Machine Learning, Artificial Intelligence, Computational Photography

Work Experience

Artisk, Software Engineer Intern | New York, NY (Remote)

Jun. 2025 – Aug. 2025

- Developed **PHP**-based **FastAdmin** dashboard for **MongoDB** data management with custom UI components and automated features
- Built a backend service with **Express.js** and **MongoDB**, providing APIs for the web frontend and admin dashboard
- Built an image layering tool with the idea of cutting and pasting in **Python** to decompose images for traditional image editing

PhinD, Software Engineer Intern | Dover, DE (Hybrid)

Jan. 2025 – May 2025

- Built a robust file management system with **AWS S3** integration, implementing real-time file preview functionality
- Developed efficient **RESTful API** endpoints with **FastAPI** to optimize frontend-backend communication
- Built **PyTorch**-powered job recommendation system for PhD students using text embeddings and lexical analysis
- Managed **PostgreSQL** database on **AWS RDS**, implementing CRUD operations via **SQLAlchemy** ORM
- Deployed backend server on **AWS EC2** with **AWS Linux**, configuring **Nginx** for enhanced security and performance

University of Illinois Urbana-Champaign, Course Assistant | Champaign, IL

Jan. 2024 – May 2025

- Provided guidance on computer graphics and **Unreal Engine** game development to 150+ students through office hours
- Evaluated programming assignments and provide comprehensive feedback to help students improve coding skills
- Served as primary technical mentor for 3 student game development teams each semester to give guidance

University of Illinois Urbana-Champaign, Library Technician | Champaign, IL

Jul. 2022 – Mar. 2023

- Resolved complex software and hardware issues, including network configuration, component compatibility
- Executed security protocols for disk imaging and secure data erasure under institutional standards
- Built computer systems with dedicated graphics processing units and configured network connectivity infrastructure

Projects

Hybrid Image with Generative AI | Champaign, IL

Sep. 2024 – Dec. 2024

- Engineered innovative image hybrid techniques using Fast Fourier Transform, creating distance-dependent visual effects
- Developed advanced image hybridization with factorized diffusion models to produce color, motion, and triple hybrid visual effects
- Designed and trained a Generative Adversarial Network optimized for real-time image hybridization to improve production efficiency

AirAware: Air Pollution Visualization Website | Champaign, IL

Sep. 2023 – Jan. 2024

- Built a **MySQL** database to store and manage 100MB daily air pollution data across the United States
- Used **React.js** to design a user-interactive web page with multiple dynamic charts for visualizing air pollution
- Engineered scalable **REST APIs** with **Express.js**, implementing efficient data handling with optimized HTTP request-response patterns
- Developed full-featured user authentication system with integrated commenting and rating functionality backed by database storage
- Used **MySQL Workbench**, **Sqlectron** and **Postman** to test the backend API connection with the database

Illini RoboMaster, member of vision & embedded team | Champaign, IL

Aug. 2021 – Oct. 2022

- Developed the auto-aiming system for the shooting system on the robot hero with a real-time object detection algorithm (YOLO v3)
- Implemented embedded systems in **C/C++** for an auto-balancing chassis with Mecanum wheels and different motors
- Achieved 2nd Place in RoboMaster University League North America (Summer 2022), competing against 10+ university teams